

REEFPRINT / KHANYA: reading the ore on the belt.

Hyperspectral hardness prediction and microscope phase identification, turned into gated, human-approved plant decisions for South African concentrators

Technical report · Mintek–SCi Grad Hackathon 2026 · Problem 3: Computer vision for real-time mineralogical characterisation

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October 2026

Abstract

South Africa mined about 71% of the world’s platinum in 2025, yet its concentrators still set feed rate, grind and reagents before the laboratory can say what was in the ore. REEFPRINT/KHANYA reads the ore at three speeds and turns what it reads into one decision. On the belt, a visible and short-wave infrared (VNIR–SWIR) hyperspectral camera feeds a model at the site that predicts the ore’s Bond work index with a one-sided 90% conformal upper bound, in 92 ms per parcel on a CPU. On 146 public drill-core composites, the model explains 48% of the variance in work index out of fold ($R^2 = 0.479$, against 0.210 for an ordinary colour camera and -0.03 for the average), and its 80% intervals cover 79.5% of samples. Setting the mill feed from the bound gave a simulated throughput gain of +1.9% (95% CI +0.9 to +2.9%), with overloads non-inferior within a pre-registered margin. In the laboratory, a reflected-light microscope layer identifies pyrrhotite, chalcopyrite and pentlandite (IoU 0.8695, 0.5755 and 0.5468 on 12 held-out sections; mean IoU 0.5725 pooled, 0.4671 per section; magnetite is not detected) and made no unsafe recommendation in 12 sections. Every proposal passes explicit gates and waits for a person to approve it before the ore reaches the mill; otherwise the site’s conservative setting applies, and the decision is recorded in a hash-chained, post-quantum-signed log. All results come from public data replayed offline: no South African belt has been scanned, and the plant is simulated. We ask Mintek and a producer for a nine-month pilot: one belt, one truth laboratory, paid on measured value.

Keywords: hyperspectral imaging; Bond work index; conformal prediction; process mineralogy; PGM concentrators; advisory process control

1 Introduction

1.1 The problem, and why it is South Africa’s

The brief states the problem plainly: “without real-time feedback, processing plants cannot adjust to changes in ore quality, leading to inefficient chemical usage and lower mineral yields”. Ore reaches the mill within minutes of leaving the belt. Assays and automated mineralogy, such as the QEM-SCAN work of Mintek’s own Minerals Processing and Characterisation Cluster [1], are batch laboratory work that reports after that ore has been processed. The plant runs three shifts a day, so each shift sets the mill and the flotation circuit for ore whose mineralogy it has not yet seen.

The cost of deciding blind falls most heavily on South Africa. In 2025 it mined an estimated 120,000 kg of the world’s 170,000 kg of platinum (about 71%) and holds about 63,000 of the world’s more than 76,000 tonnes of PGM reserves (about 83%), mostly in the Bushveld Complex [2]. The same source attributes the 9% fall in South African PGM output in 2025 to lower palladium prices, higher deep-level mining costs and ongoing disruptions to the electricity supply. Zimbabwe, the third-largest platinum producer (18,000 kg in 2025), mines the same family of ores on the Great Dyke [2].

Recovery is where the money is. UG2 ore carries about 30% Cr_2O_3 against about 0.1% for Merensky ore, and the smelter caps the chromite in concentrate. Mintek testwork

made UG2 concentrate at 87% recovery with 2.9% Cr_2O_3 , and showed that grades and recoveries above 90% were possible only if the Cr_2O_3 limit was relaxed [3]. Producers report recovery moves of this size to investors: Valterra improved concentrator recoveries by 1–2 percentage points at Mototolo and Amandelbult in 2025 [4]. Comminution, which sets liberation and mill throughput, used 52% of site electricity in a benchmark of gold and copper mines [5], on a South African grid that is itself a constraint.

1.2 What existing instruments cannot tell the shift

Plants already run many on-line sensors (Table 1). Each measures one property in one place. Advanced process control, including Mintek’s MillStar and FloatStar, stabilises the circuit from process measurements. What is missing is an *ore-side, uncertainty-aware input* to that layer: one that arrives before the ore reaches the mill, states how wrong it might be, and refuses to advise when its evidence is thin. A belt-mounted hyperspectral camera can supply part of that input. It carries no radioactive source, unlike isotope-source neutron-activation analysers, and it sees the ore before the mill does. It does not see sulphide minerals or PGMs, which is why a laboratory truth layer stays in the design.

Table 1: Plant instruments and what each cannot give the shift. No performance figures are implied.

Instrument	Measures	Cannot give the shift
Cross-belt PGNAA/PFTNA	Bulk elements, minute by minute	Minerals, hardness, PGM at g/t
On-stream XRF (slurry)	Cu, Ni, Fe, Cr in process streams	Trace PGM; ore before the mill
Froth and size cameras	Froth state; rock size	The ore’s mineralogy and hardness
Automated mineralogy (QEMSCAN, MLA)	Phases, liberation, association	Results on the shift’s timescale
APC (e.g. MillStar, FloatStar)	Stabilises levels, flows, mill load	A feed-forward ore input
Belt hyperspectral (this work)	Silicate, clay, carbonate and iron-oxide absorption features	Sulphides and PGMs (opaque in the SWIR)

Table 2: The brief’s requirements and where this report evidences them.

Brief requirement	Evidence
≥ 3 mineral phases	Microscope: pyrrhotite, chalcopyrite, pentlandite (Sec. 4.4)
Accuracy report	Belt: R^2 , interval coverage, baselines (Sec. 4.1). Microscope: per-class, per-section, decision-level (Sec. 4.4)
Adjust a plant parameter	Belt bound $\rightarrow$ mill feed rate; microscope advice $\rightarrow$ regrind, over OPC UA to a simulator
Real-time	92 ms per belt parcel; about 3 s per microscope field (CPU)
Integrates with controls	Advisory input inside site limits; OPC UA; designed to feed APC

1.3 Contribution and scope

The system reads the ore at three speeds and turns all three into one decision (Section 3):

1. **the belt** (REEFPRIENT): a hyperspectral model predicts each parcel’s grinding hardness with an upper bound, and Bond’s law turns the bound into a mill feed-rate proposal;
2. **the laboratory** (KHANYA): a reflected-light microscope model identifies the sulphide phases and their liberation, the brief’s three-phase deliverable;
3. **the decision**: a person approves each proposal within the ore’s transit time, or the site’s conservative setting applies; every step is signed and recorded.

Table 2 maps the brief’s requirements to the evidence. Not claimed: performance on South African ore, any measured plant benefit, mineral identification by the belt camera, or identification of chromite and the silicates by the microscope.

2 Materials and methods

2.1 Data

No South African belt scans or labelled Bushveld micrographs were available, so every model was trained and tested on public data (Table 3). HIDSAG supplies VNIR–SWIR scans of Chilean porphyry Cu–Mo drill core with laboratory Bond work-index values [6]. LumenStone S2, a Norilsk Ni–Cu sulphide ore, is an *assemblage* analogue for the base-

metal sulphides that carry PGMs in Bushveld ores, not an abundance analogue: sulphides cover 62.8% of S2 by area, against under 1 vol% in UG2 [7].

2.2 Belt layer: hyperspectral hardness

Sensor (design). The design assumes a Specim SX25-class VNIR–SWIR line scanner above the belt [9]. At 162 lines per second on a belt running at 2 m/s, the camera records a line every 12.3 mm, with 1.88 mm pixels across a 1.2 m belt. It therefore averages over parcels of ore; it does not map individual particles.

Model. Each parcel’s spectra are resampled to fixed VNIR and SWIR grids and reduced to features by spectral region, including the OH/H₂O, Al–OH and Mg–OH/CO₃ absorption regions (Fig. 2). For each target, a model family (linear or tree ensemble) is chosen by nested cross-validation within the training folds. All results are out of fold. Targets are kept only if they beat their strongest baseline under gates written before each run.

Bound and gate. The feed rate is set from a one-sided 90% split-conformal upper bound on W_i [10], not from the point prediction. An approximate CV+ bound [11] was tried first and failed its gate; the exact split-conformal bound, the plan’s stated fallback, was then used, so the result is provisional. A distance gate calibrated on the training folds flags parcels unlike anything in training. Flagged parcels receive the conservative setting.

2.3 Decision layer: from a bound to a plant setting

A ball mill is usually limited by motor power. Bond’s law gives the specific energy [12]

$$W = 10 W_i \left(\frac{1}{\sqrt{P_{80}}} - \frac{1}{\sqrt{F_{80}}} \right) \text{ kWh/t}, \quad (1)$$

so at a fixed grind target the achievable feed rate is proportional to $1/W_i$. The proposed feed is the design feed scaled by (design hardness) / (upper bound on W_i). Power, feed size and grind target cancel, so no plant parameter is assumed. The proposal must sit inside a site-approved operating envelope; the demonstration envelope is 85–110% of design feed, with 100% as the conservative setting and ramp limits of $-10\%/+3\%$ per parcel (feed is cut fast and raised slowly). These values are stipulated for the demonstration; a site replaces every one of them.

A metallurgist may approve, modify within the envelope, reject or escalate. If nobody approves before the parcel reaches the mill (an assumed 90 s transit), or the parcel is flagged, or the data are stale, the conservative setting applies and the case is escalated. The system never outputs “unknown” and never holds the last value. Every event is appended to a hash-chained record with Ed25519 and ML-DSA-65 signed checkpoints [13]. The microscope’s advice reaches a simulated regrind parameter over OPC UA [14]; the same tag map is designed for the feed rate.

2.4 Laboratory layer: microscope phases

The microscope model is torchvision’s DeepLabV3 with a ResNet-50 backbone [15, 16], trained for five classes on

Table 3: Datasets. All are public; none was supplied by Mintek or a producer. Splits are the publisher’s where one exists.

Dataset	Ore and locality	Size and split	Used for	Licence
HIDSAG GEOMET [6]	Porphyry Cu–Mo, Chile	146 drill-core composites, VNIR (39 bands, 410–980 nm) and SWIR (75 bands, 1010–2490 nm), with Bond work index	Belt hardness model; feed-rate replay	CC0
LumenStone S2 [7]	Ni–Cu sulphide, Norilsk	49 reflected-light images, 3396×2547 px; 37 train (31 + 6 validation), 12 held-out test; 5 classes	Microscope phase segmentation	Research use with citation
LumenStone S1 [7]	Hydrothermal, Berezovskoe	7 classes; the 16 test images of the published benchmark	Published comparison (App. A)	Research use with citation
Bushveld assays [8]	Bushveld Complex chromitites	1,112 intervals; 123 projects held out by project	Negative-result check on PGE routing	CC BY 4.0

8 epochs of 64 native-resolution 512px patches (batch 2, AdamW at 2×10^{-4} , cross-entropy; class-uniform patch sampling). The checkpoint under report has SHA-256 `de7135a9...`; the software refuses to command the simulator with any other checkpoint. A *Quick* scan classifies six non-overlapping 512px fields (18.2% of an S2 section); a *full-section* scan uses overlapping tiles.

Connected regions of the predicted mask are split into grains. For S2, chalcopyrite and pentlandite are *payload* and pyrrhotite is *reject*. Each grain is *free* if at least 50% of its classified area is payload, *locked* if less, or carries no valuable mineral; the 50% floor follows the fall in composite-particle flotation recovery below about half surface exposure [17]. A rule-based advisor turns these measures into *grind finer*, *continue* or *verify before acting*. Advice reaches the simulator only if the checkpoint is approved, the image is a validated sample that passes a colour-cast check, at least 9 payload-bearing particles were found, and mean confidence is at least 0.85. Three of the advisor’s four thresholds are placeholders awaiting values from site metallurgists.

2.5 Evaluation protocol

Belt models are scored out of fold against the strongest baseline (fold mean, fold median, and a model using only RGB colour), with gates fixed in a pre-registration committed before each run. The microscope uses the publisher’s 12 held-out S2 sections, never used for training or selection. Its IoU is reported pooled over all 103,795,344 test pixels and averaged per section, with percentile bootstraps over sections (2,000 resamples, seed 42). Decision-level agreement runs the advisor on the model’s mask and on the expert mask. A disagreement is *unsafe* if the model says *continue* where the expert-mask advice does not, or gives non-hedged advice where the expert mask says *grind finer*.

3 System architecture

Figure 1 and Table 4 summarise the system. The belt monitor (REEFPRINT Live) runs the hardness model, the decision view and the signed record. The microscope workbench (KHANYA) is a React and TypeScript application on a FastAPI service with PyTorch inference on the CPU; in local mode it accepts only loopback connections and makes no external network requests. Both run offline on one laptop, so a power or network outage does not stop the advice. An op-

Table 4: Three speeds, one decision. Speeds are measured on a CPU except the approval window, which is an assumed belt-to-mill transit.

Layer	Input	Output	Speed
Belt (REEF-PRINT)	VNIR–SWIR spectra per parcel	W_i with a 90% upper bound; out-of-distribution flag	92 ms per parcel
Laboratory (KHANYA)	Reflected-light micrograph of a polished section	Phases, grains, liberation, advice	16.7–21.2 s per six-field scan
Decision	Both outputs and the site envelope	Feed-rate or regrind proposal, approved or conservative	90 s window (assumed)

tional assistant may route questions, but no language model computes a measured or control value.

Three environments exist, and each claim names its environment. A static public demonstration runs the belt monitor on precomputed data, with an unsigned browser-sandbox record. The secure server (role-based access, scrypt password hashing, CSRF protection, rate limits, signed ledger) passed 17 of 17 security tests locally and has not been deployed publicly. The microscope workbench ran locally. **Software verification:** the workbench has 180 pytest functions and 7 browser tests, with four continuous-integration checks passing; the belt branch has 328 pytest functions, and static analysis (Bandit) reported 0 issues after a dependency audit led to an upgrade of cryptography.

4 Results

4.1 Belt: predicting grinding hardness

Of the five HIDSAG targets tested (copper and molybdenum recovery, lime consumption, pH and Bond work index), only the work index beats its strongest baseline under every gate. Table 5 gives the out-of-fold results. The hyperspectral model explains 48% of the variance; a model given only RGB colour explains 21%, so the infrared bands carry most of the information. The 80% intervals cover 79.5% of samples, as intended.

Figure 2 shows one parcel as the model sees it. For parcel GMET-0004, the predicted W_i is 15.9 kWh/t and the one-sided 90% bound is 16.9 kWh/t, against a design hardness

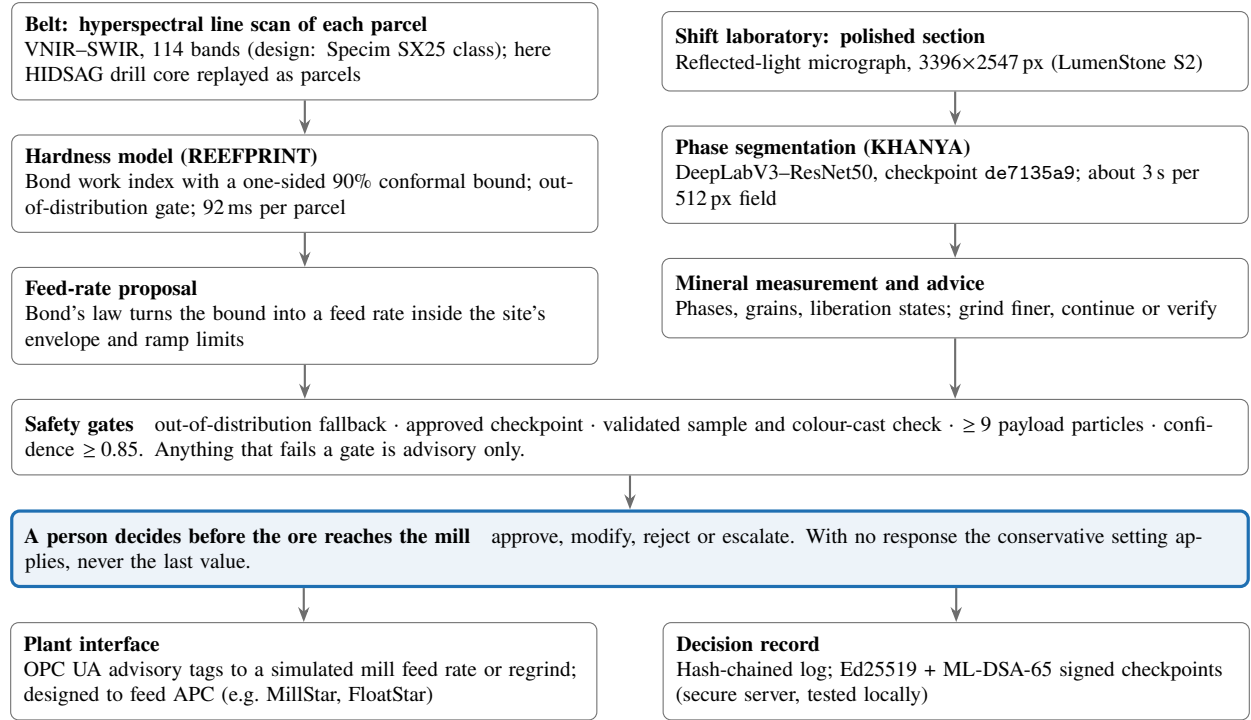


Figure 1: System architecture. The belt layer (left) is the site system; the microscope layer (right) is the laboratory truth layer and delivers the brief’s three phases. Both pass the same gates and the same human decision. Everything runs offline on one laptop CPU. The plant is a simulator: no plant data or plant connection exists.

Table 5: Bond work index, 146 HIDSAG composites, out of fold. Mean measured W_i 14.62 kWh/t.

Predictor	R^2	MAE (kWh/t)
Fold mean (no ore information)	−0.027	1.463
Fold median	−0.041	1.471
RGB colour only	0.210	1.291
VNIR–SWIR hyperspectral	0.479	1.052
80% interval coverage: 0.795 (target 0.80)		

of 16.6 kWh/t (the training folds’ 90th percentile). The proposal is therefore 98.2% of design feed. The laboratory value for this composite is 12.1 kWh/t: the bound was conservative here, which is the direction a safety bound should err.

4.2 Belt: from hardness to a plant decision

Figure 3 shows the decision for the same parcel as the operator sees it, and Table 6 replays four feed-rate policies over all 146 composites. Setting the feed from the bound, with the out-of-distribution fallback, gave a simulated throughput gain of +1.9% (95% CI +0.9 to +2.9%). Overloaded parcels fell from 8.9% to 6.8%; the one-sided 95% upper bound on the difference is +1.4 pp, inside the pre-registered +3 pp non-inferiority margin. Feeding from the point prediction instead would overload almost half the parcels, which is why the policy uses a bound. The policy captures 13% of the gain that perfect information would give, so better hardness models have room to add value.

The result is provisional: the exact bound was adopted after the approximate bound failed, HIDSAG has no time order (so stale data, transit time, stockpile mixing and ramp limits are not simulated), and the ore is not Bushveld ore. A robustness test, pre-registered before it was run, found that a new

Table 6: Simulated feed-rate policies on 146 HIDSAG composites (not PGM ore, not a time series). Throughput is relative to no ore information.

Policy	Throughput (95% CI)	Overloaded
No ore information (90th-percentile hardness)	0 (reference)	8.9%
Belt, one-sided bound + fallback	+1.9% [+0.9, +2.9]	6.8%
Belt, point prediction	+13.6% [+12.3, +14.8]	45.2%
Perfect information (ceiling)	+14.0% [+11.7, +16.4]	0%

capture of the same ore keeps its error (mean absolute error $\times 1.00$, 95% CI 0.99–1.01). Half the field of view ($\times 1.01$), added noise ($\times 1.00$) and $\pm 15\%$ illumination ($\times 1.01$) were also tolerated. A $\times 1.25$ wavelength drift went mostly unflagged, so a wavelength-calibration gate was added to the hardware design.

4.3 Is it physically possible?

Every claim was tested against its physics, and the failures stay in the report (Table 7). The most useful negative check concerns load-shedding: re-ordering ore during electricity curtailment cannot create tonnes, because energy per tonne is conserved over a cycle, so that claim was rejected.

4.4 Laboratory: phase identification

The microscope meets the brief’s three-phase requirement (Table 8). Magnetite, a fourth class the brief does not require, is never predicted: 78.3% of it is labelled as the dark mounting resin (Appendix A). Pixel accuracy is 0.8914, but background and pyrrhotite dominate it, so it is not a measure of mineral identification.

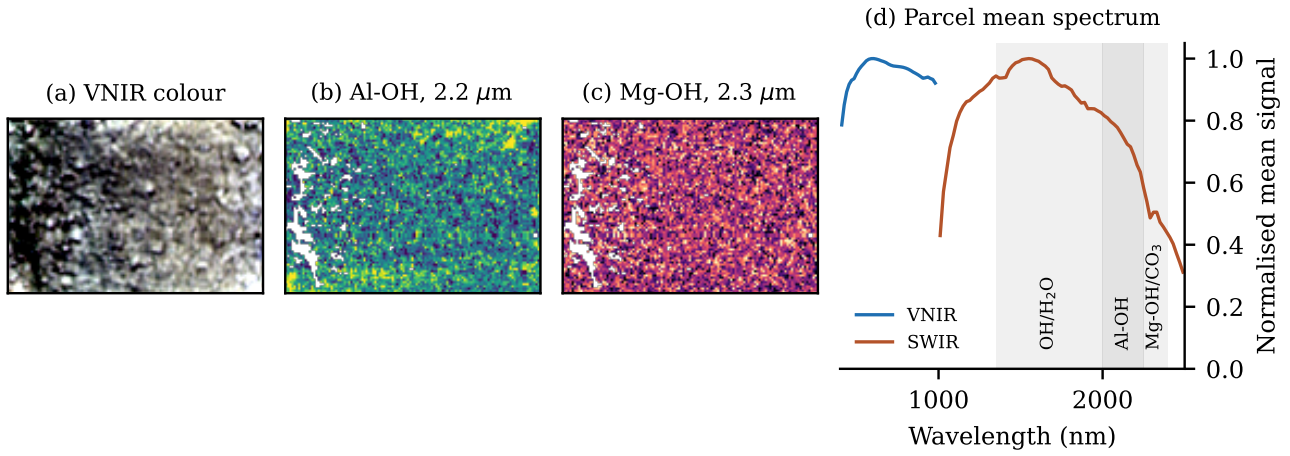


Figure 2: One belt parcel as the hardness model sees it: HIDSAG drill-core composite GMET-0004 (Chilean porphyry Cu–Mo), 80×117 pixels. (a) Colour composite from the VNIR bands. (b, c) Relative absorption depth at the Al–OH (white mica, clays) and Mg–OH (chlorite, biotite, carbonate) features; detector column offsets removed for display only, as in the application. (d) Parcel mean spectrum per sensor, with the absorption regions the model uses. From this scan the model predicts W_i 15.9 kWh/t with a 90% upper bound of 16.9 kWh/t (measured: 12.1 kWh/t). The maps show absorption features, not a mineral identification.

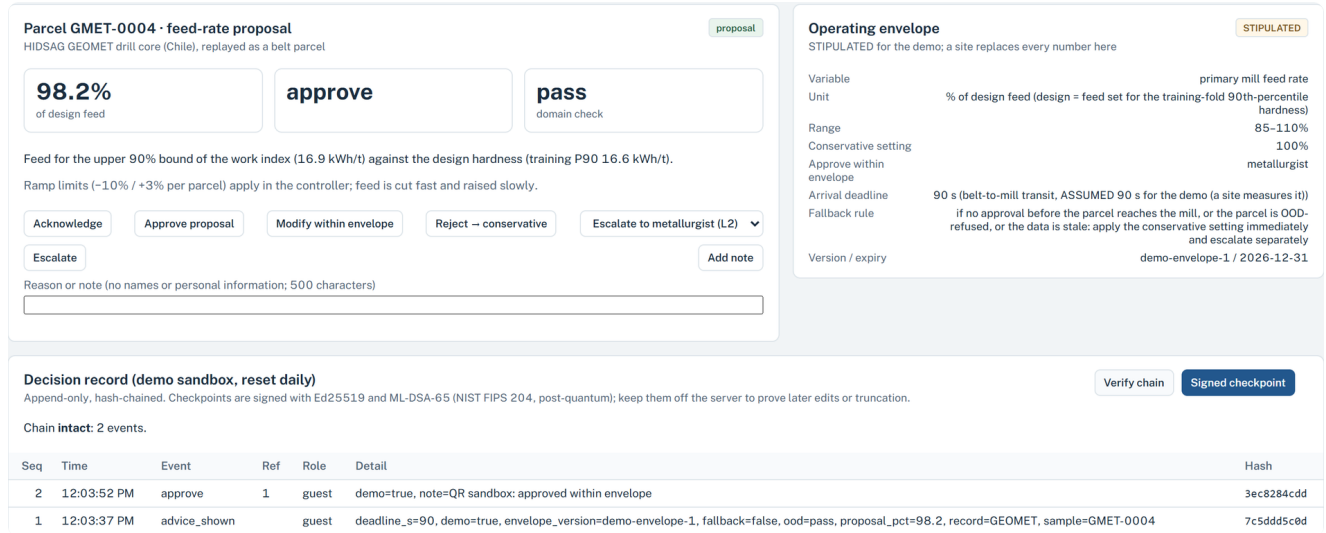


Figure 3: The Decisions view of REEFPRINT Live for parcel GMET-0004: a feed-rate proposal of 98.2% of design inside a stipulated demonstration envelope, the operator’s actions, and the hash-chained decision record with a guest approval (demonstration sandbox). Screenshot of the application.

The pooled mean IoU is 0.5725 (95% CI 0.494–0.624); averaged per section it is 0.4671 (95% CI 0.412–0.534), and 10 of the 12 sections fall below the pooled figure (Fig. 4). The model clearly beats trivial baselines (majority class 0.1167; colour-only nearest centroid 0.3948). A second run of the same recipe scored 0.4543 because the patch sampler is unseeded, so these figures describe this checkpoint, not the method. A retrained candidate scores 0.6320 and detects magnetite, but loses pyrrhotite (0.8183) and pentlandite (0.4460), the phases that matter most for PGM deportment, so it is not deployed. On the published Lumen-Stone S1 benchmark our separate S1 model scores 0.7224 against 0.8506 [18].

4.5 Laboratory: decisions and grain evidence

Running the advisor on the model’s masks and on the expert masks, the shipped (topology-refined) pipeline disagrees on 8 of 12 sections, and every disagreement is a request for veri-

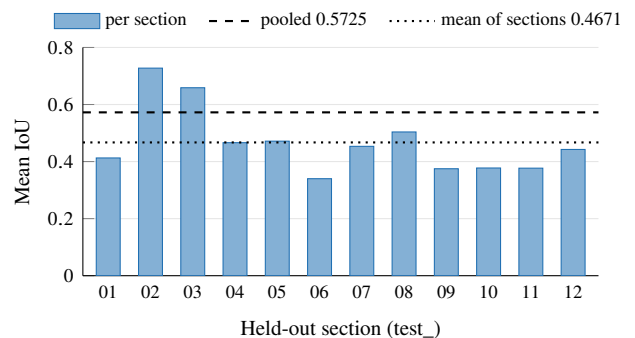


Figure 4: Microscope: mean IoU of each held-out section. Ten of 12 sections fall below the pooled figure; the range is 0.340–0.728.

fication: 0 unsafe and 0 conservative. The raw pipeline made 2 unsafe errors. Zero unsafe in 12 is an observation, not a rate; the one-sided 95% upper bound on the unsafe rate is 22.1%. A six-field scan matches the whole-section advice

Table 7: Physical-possibility checks. Modelled values are labelled; none is a plant measurement.

Claim	Result	Status
Hardness bound gains tonnes (Bond)	+1.9% [+0.9, +2.9], simulated	Provisional
Without losing flotation recovery	Modelled cost ≤ 0.25 pp on the steep part of the curve; 0 on the plateau	Check residence first
More tonnes under curtailment by re-ordering ore	-3.8% simulated; even perfect knowledge gives -3.0%	Rejected
Works on an unseen capture	Error $\times 1.00$ [0.99, 1.01]	Holds
Real-time	92 ms median per parcel, 4-core CPU	Measured
Camera resolves the belt	12.3 mm lines, 1.88 mm pixels: parcel-level, not particle-level	Computed
Belt camera identifies ≥ 3 minerals	No pre-registered spectral hypothesis passed; the microscope carries the phases	Not shown
Detects radiation	No: optical only; a gamma monitor can be added	No

Table 8: Held-out results for checkpoint de7135a9, 12 Lumen-Stone S2 sections, full-section inference.

Class	IoU	IoU (void-border)	Recall	Precision
Background (resin)	0.8709	0.9290	0.9198	0.9425
Pyrrhotite	0.8695	0.8929	0.9343	0.9261
Chalcopyrite	0.5755	0.6151	0.7161	0.7456
Pentlandite	0.5468	0.5801	0.7420	0.6752
Magnetite	0.0000	0.0000	0.0000	–
Mean (5 classes)	0.5725	0.6034		

on 9 of 12 sections, against 0 of 12 for a single centre field. A realistic exposure shift (–35 RGB levels) changed the advice on 8 of 12 sections, so illumination control is a pilot requirement.

The grain inspector (Fig. 5) lets a mineralogist audit the advice grain by grain: grain 30 on section test_11 is 76.5% pyrrhotite and 23.5% pentlandite, so it is *locked*, while grain 31 beside it is pure pentlandite and *free*. In Quick mode each field is classified without context beyond its edge; on test_11, 19 of 24 grains (8 of the 10 carrying valuable mineral) come within 6 px of a field edge, so Quick-mode grain statistics are indicative and the full-section scan is the reference. Latency on a laptop CPU: 2.64 s mean per 512 px field ($n = 60$); 16.7–21.2 s for a six-field scan on mains power (5 runs) and 30.8 s on battery; 162.3 s for a whole section.

4.6 Negative results

Table 9 lists results the team tested and withdrew. They mark where the method stops.

5 Discussion

5.1 What changes on a shift

Two levers are covered, each with a named owner. The control room receives a hardness bound and a feed-rate proposal for each parcel before it reaches the mill. The metallurgist receives liberation evidence from the laboratory for grind

Table 9: Negative and withdrawn results.

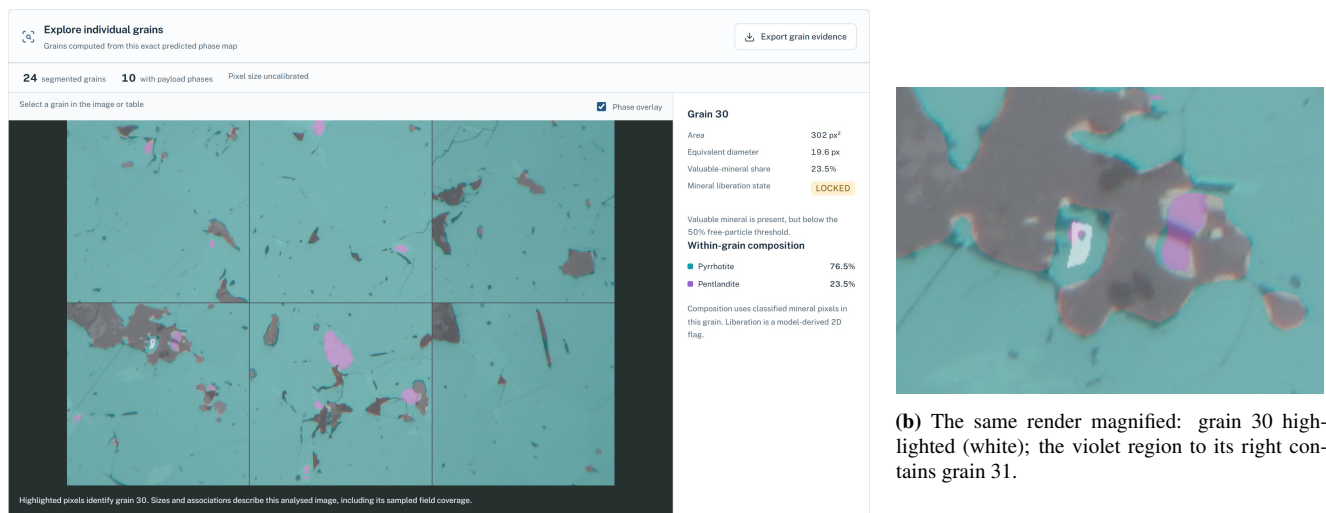
Test	Result
Belt spectra predict QEMSCAN mineralogy	Withdrawn: size fraction and process line explain it; spectra add skill on 0 of 33 minerals
Belt chemistry routes Bushveld PGE ore	The mine-plan seam routes better: balanced accuracy 0.82 vs 0.79
Belt predicts recovery, lime, pH	No significant gain over the strongest baseline
Plant tags forecast concentrate silica 1 h ahead	Never beats the last laboratory assay
Wider model menus; seed ensembles	No gain past the pre-registered gates

and regrind decisions, and can audit it grain by grain. In both cases the system is an advisory input inside limits the site sets, beside existing process control such as MillStar and FloatStar; it is not a controller. Two design choices carry the safety argument: a bound rather than a point prediction (Table 6), and a conservative fallback rather than the last value, because the last value may itself be wrong.

5.2 Why this matters to Mintek, and where it can go

Mintek’s mandate is “to promote mineral technology and foster the establishment and expansion of industries in the field of minerals and their derived products through research, development, and technology transfer” [19]. The system fits that mandate in four ways, each drawn from Mintek’s own published work.

- 1. More truth-laboratory work, not less.** The models need QEMSCAN composites and Bond tests to retrain and audit them. Mintek’s Minerals Processing and Characterisation Cluster runs QEMSCAN [1]; Mintek already characterises drill core mineralogically for chromite and iron-ore washability, and applies ore characterisation to flotation reagent selection for PGMs and base-metal sulphides [19].
- 2. A feed-forward input to Mintek’s process control.** Mintek reports a 0.5% throughput gain at a lead–zinc concentrator from AI-driven steady-state optimisation, worth over US\$2 million a year [19]. An ore-hardness bound that arrives before the ore is the kind of input such optimisation can use.
- 3. A business model Mintek has already proven.** Mintek’s Process Control Implementation team’s benefit-sharing model with Hindustan Zinc (Vedanta) in India links revenue to system performance and has generated over R35 million since 2021 [19]. Our proposed pilot is priced the same way: paid on measured value.
- 4. A route beyond South Africa.** Mintek’s products and services are used in over 60 countries, its instruments group operates in 42, and it works through distributors including Molycop and Mercantil [19]. Its footprint map includes Zimbabwe, the second PGM province of the region, and Zambia, the DRC and Chile, major copper-producing countries; the belt model was developed on Chilean copper ore. Mintek’s flotation group aims to become a globally recognised centre of excellence in flotation research through technology partnership [20].



(a) Grain inspector with grain 30 selected.

Figure 5: Laboratory layer: the grain inspector on held-out section test_11 (Quick scan, checkpoint de7135a9). Grain 30 (302 px²): 76.5% pyrrhotite, 23.5% pentlandite, locked under the 50% rule; grain 31 beside it is 100% pentlandite and free. Pixel size is uncalibrated. Screenshot of the application.

5.3 Scope and limitations

1. **No South African ore has been evaluated.** The belt data are Chilean Cu–Mo core and the microscope data are Norilsk ore.
2. The belt result is simulated on 146 composites replayed as parcels, without time order; it is provisional until a pilot measures it.
3. The belt camera does not identify minerals; it predicts hardness. Chromite and the silicates have not been trained on by the microscope, and magnetite is never detected.
4. The 12 held-out microscope sections were reused during development and may not be independent specimens; the training recipe is not reproducible.
5. The microscope’s advice is sensitive to illumination, and three advisor thresholds are unsourced.
6. Liberation is measured in 2D and is an upper bound; in Quick mode most grains are cut by field borders.
7. Latency is CPU-only with small n . The plant interaction is a simulation; no claim is made about recovery, reagents or turnaround at a plant.

Table 10 compares the build with the abstract Mintek accepted. Most differences follow from two facts: no Bushveld image, belt or QEMSCAN data was available, and the team built software only.

5.4 Indicative value

No plant benefit has been measured, so value is shown as a worked example with labelled assumptions (Appendix C). Northam’s Zondereinde mine produced 333,050 equivalent refined 4E ounces in F2026 [21]. One per cent more recovered metal at that output is 3,331 ounces a year. At Valterra’s realised basket prices of R32,611 per PGM ounce (2025 average) and R45,993 (first half of 2026) [4, 22], that is R108.6 M to R153.2 M a year in revenue, before any incremental processing or refining cost. Against an assumed belt installation of US\$0.5–3.0 M with US\$0.1–0.6 M a year in running costs (five years at 10%, R17.61 per US\$), the pilot breaks even at

0.03–0.23% more recovered metal. These are scenarios, not forecasts; the pilot exists to measure the real number.

6 The ask and the roadmap to a pilot

The ask: one belt, one truth laboratory, nine months, paid on measured value. We ask a South African PGM producer to host a pilot on one belt, and Mintek to act as its truth laboratory. The pilot is built so that every decision to spend more is taken on measured evidence, and so that a “no gain” result is reached early and cheaply.

6.1 Who provides what, and what each receives

Table 11 sets out the four partners. The producer and Mintek each receive something they can use during the pilot, not only at its end, and the team carries the commercial risk: no measured value, no fee.

6.2 Roadmap with go/no-go gates

Figure 6 and Table 12 set out the nine months. Each phase ends at a gate that the partners agree in advance. If a gate fails, the pilot stops and the spend to that point is the producer’s maximum exposure.

6.3 The investment case

The pilot needs to find a small effect to pay back. At Zondereinde’s F2026 output, each extra 1% of recovered metal is worth R108.6–153.2 M a year in revenue (Section 5.4). Across an assumed installed cost of US\$0.5–3.0 M and running cost of US\$0.1–0.6 M a year, break-even is 0.03–0.23% more recovered metal. For scale, Valterra reported concentrator recovery gains of 1–2 percentage points in 2025 [4], and Mintek’s own AI optimisation raised a concentrator’s throughput by 0.5%, worth over US\$2 million a year [19].

Table 10: Proposed in the accepted abstract versus delivered.

Proposed	Delivered	Status
Five Bushveld phases (chromite, orthopyroxene, plagioclase, base-metal sulphide, talc)	Pyrrhotite, chalcopyrite, pentlandite on a Norilsk analogue; magnetite fails	Changed: no labelled Bushveld data
QEMSCAN maps as “teacher” for an optical “student”	Not built; becomes the pilot’s truth-laboratory stream	Deferred to the pilot
Grindability via Bond work index	From belt hyperspectral data: R^2 0.479, bounded, decision-grade	Delivered, as the core result
Liberation and texture	2D grains, liberation states, contacts, liberation by size	Delivered (2D)
Conformal intervals, out-of-distribution gate	On the belt, with a pre-registered robustness test	Delivered
Abstain by holding the last-known-good setpoint	Abstain by applying the site’s conservative setting	Deliberately reversed
OPC UA feed-forward, advisory namespace outside the safety system	OPC UA to a simulated parameter; human approval; signed record	Delivered (simulated plant)
Edge inference (Raspberry Pi 5 + Hailo-8L)	Laptop CPU: 92 ms per parcel	Not done; the CPU meets the belt rate

Table 11: The ask. Costs are hypotheses or assumptions until quoted; target producers are named as segments, not customers (e.g. Valterra, Implats, Sibanye-Stillwater, Northam, Tharisa).

Partner	Provides	Receives	Cost
Producer (general manager, metallurgy)	One belt with camera access; laboratory assays and Bond tests it already runs; read-only plant-historian data	Hardness bounds and liberation evidence every shift; a measured on/off result on its own plant; after the pilot, pays only a share of measured value	Pilot: US\$0.5–3.0 M installed + US\$0.1–0.6 M a year (assumed); then at most 20% of measured net value
Mintek	Truth laboratory (QEMSCAN composites, Bond tests); process-control expertise and the route to MillStar and FloatStar; technology-transfer framework	Paid laboratory work; a feed-forward ore input for its process control; a performance-linked product like its benefit-sharing model [19]	Laboratory work billed to the pilot
Development funder (e.g. the Technology Innovation Agency)	Bridge funding for agreement, installation and the laboratory pilot	A South African-owned technology taken from TRL 4 to TRL 6–7	Lab pilot about R210,000 (hypothesis)
Team Sonar	Software, models, integration, analysis and reporting; the risk of no fee if no value is measured	Licence of about R12,000 a month per site after the laboratory pilot (hypothesis)	–

The break-even is a small fraction of moves the industry already reports.

The commercial structure keeps the producer’s risk bounded. Its maximum exposure is the pilot’s installed and running cost, with three exits on the way. After the pilot, the fee is a share of measured value, so it is zero if no value is measured. Mintek already earns this way: its benefit-sharing model with Hindustan Zinc has generated over R35 million since 2021 [19].

6.4 Risks, and how the pilot retires them

Table 13 lists the main risks an executive should weigh. Each is either tested at a named gate or limited by design, so none is left for the producer to discover after a full roll-out.

6.5 Technical work inside the pilot

The technical work follows from the limitations: retrain and validate on QEMSCAN-labelled UG2 and Merensky material with locality-disjoint tests; add chromite and the silicates; test the belt layer on Bushveld core, guided by published hyperspectral work on Merensky drill core [VERIFY: read and cite the 2025 Minerals Engineering study]; add colour normalisation and a wavelength-calibration gate; set the advisor’s thresholds with Mintek metallurgists; and use the liberation classes of Moodley et al. [1] to design the next

timed-concentrate test.

7 Conclusions

A hyperspectral camera on the belt can predict grinding hardness well enough to steer a mill: on 146 public drill-core composites the model explained 48% of the variance in Bond work index, more than twice what colour alone explains, and a feed-rate policy built on its upper bound gave a provisional simulated throughput gain of +1.9% without more overloads. A reflected-light microscope model identified three sulphide phases on held-out sections, with magnetite as a stated failure, and gave no unsafe advice in 12 sections. Both feed one decision that a person approves and a signed record keeps. These results hold for public analogue ores and a simulated plant. Whether they transfer to South African ore is the question a nine-month pilot, with Mintek as the truth laboratory, is designed to answer.

CRedit authorship contribution statement

Lethabo Hoacane: Conceptualisation, domain lead, belt hyperspectral modelling, belt monitor and decision application, economics, presentation. **Sibusiso Khumalo:** Microscope segmentation model, accuracy and decision-level evaluation, advisor and safety gates, OPC UA simulation, workbench review and verification, technical report. [VERIFY: Lethabo to confirm these roles]

Declaration of competing interest

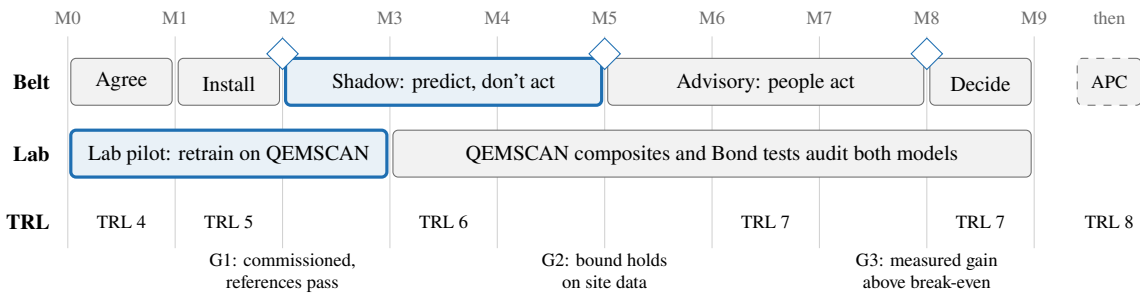


Figure 6: Roadmap to a decision in nine months. Agree covers data sharing, POPIA and operational-technology access. In shadow mode the system predicts but nobody acts; in advisory mode people approve proposals and on/off blocks measure the effect. Diamonds are go/no-go gates (Table 12). TRL levels are the team’s self-assessment.

Table 12: What the partners get at each phase, and the proposed test to continue. Criteria are proposals for the partners to agree before the pilot starts.

Phase	Months	What the partners get	Go/no-go test to continue
Agree	0–1	Data-sharing, POPIA and OT-security agreements; the site’s operating envelope and conservative setting, set by its metallurgist	Agreements signed; envelope approved
Laboratory pilot	0–3	Microscope retrained on QEMSCAN-labelled sections of this ore; measured agreement with QEMSCAN and turnaround	Beats the current model on held-out localities; agreement with QEMSCAN reported
Install and commission	1–2	Camera, white and wavelength references, edge PC, read-only advisory tags	G1: references and calibration pass; no write path to any safety system
Shadow	2–5	Hardness bounds logged against the site’s Bond tests; nobody acts on them	G2: bound coverage holds on site data; out-of-distribution rate acceptable. Otherwise retrain or stop
Advisory	5–8	People approve proposals; on/off blocks compare throughput, overloads and recovery with current practice	Overloads non-inferior; effect estimated with a confidence interval
Decide	8–9	A measured-value report to the producer and Mintek	G3: gain above break-even → subscription and coupling to APC; otherwise stop, and no fee is due

Table 13: Main risks and how the pilot tests each one.

Risk	How the pilot tests or limits it
The hardness model does not transfer to Bushveld ore	Shadow phase compares bounds with site Bond tests (G2); retrain on site data or stop at month 5
Dust, moisture, vibration, lighting and wavelength drift on the belt	Reference targets and a calibration gate at commissioning (G1); drifted parcels are flagged and get the conservative setting
Power outages and load-shedding	Runs offline on an edge PC with a UPS; after a restart the conservative setting holds until checks pass
Cyber security and plant safety	Read-only advisory tags, no write path to safety systems, signed records; site IT/OT sign-off before installation
Operators ignore or over-trust the advice	Named approval roles, a fixed approval window, escalation; measured in the advisory phase
No measurable gain	On/off blocks with confidence intervals; stop at G3, and no fee is due

The authors declare no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We thank the Mintek mentors (a chemist, a mineralogist and a metallurgist) for a review session that reshaped the build **[VERIFY: names, with their permission]**, and the authors of HIDSAG, LumenStone and the Bushveld assay dataset for releasing their data.

Use of AI tools

Code and drafting were assisted by AI coding assistants. Every re-

ported number was computed by code in the project repository from the files named in Appendix B, and was checked by the authors.

Data and code availability

HIDSAG is available under CC0 (doi:10.6084/m9.figshare.c.5983921) [6]; LumenStone under research-use terms [7]; the Bushveld assays under CC BY 4.0 (doi:10.17632/dc8jcnbev.1) [8]. Code, checkpoint SHA-256 values and evaluation outputs are in the team repository (Sibusiso-K/KHANYA); access is available to Mintek on request under the hackathon’s IP terms **[VERIFY: confirm with the team]**.

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A Additional microscope results

Table 14: Row-normalised confusion matrix (% of each true class), all 103,795,344 S2 test pixels.

Truth \ pred.	Bkg	Ccp	Mag	Po	Pn
Background	92.0	0.8	0.0	5.7	1.5
Chalcopyrite	1.6	71.6	0.0	3.1	23.7
Magnetite	78.3	0.1	0.0	16.8	4.8
Pyrrhotite	1.1	1.6	0.0	93.4	3.9
Pentlandite	0.7	0.8	0.0	24.3	74.2

Table 15: Void-border IoU on the 16 test images of the published S1 benchmark (byte-identical, no overlap with our training data).

Class	KHANYA (S1)	Published	Gap
Bornite	0.8975	0.8955	+0.0020
Background	0.8490	0.8505	-0.0015
Pyrite	0.9286	0.9732	-0.0446
Chalcopyrite	0.8916	0.9363	-0.0447
Sphalerite	0.6355	0.7653	-0.1298
Galena	0.5068	0.7630	-0.2562
Tennantite	0.3479	0.7706	-0.4227
Mean	0.7224	0.8506	-0.1282

B Verification and reproduction

Sources of every result (repository Sibusiso-K/KHANYA):

- *Belt*, branch codex/pwa-phase-roadmap (ab33307):
training/hidsag-v6-live-20261001/output/hidsag_v6_results.json (Table 5, recomputed from the stored out-of-fold predictions), training/value-chain-20261002/results.json and docs/16-decision-value-chain.md (Table 6), training/hidsag-v9-robustness-20261002/output/, training/physics-checks-20261002/results.json, training/installation-20261002/sensor_geometry.json (Table 7), presentation/belt-monitor/live/showcase/GEOMET/GMET-0004.* (Fig. 2).
- *Microscope*, branch codex/launch-live-demo: reports/ACCURACY-REPORT.md, lumenstone_s2_patches_test_metrics.json, s2_section_stats.json, magnetite_confusion_patches.json, lumenstone_s2_trivial_baselines.json, decision_gap_patches_refined.json, field_sampling_s2.json, segmentation_latency.json, benchmark_s1_v1_protocol.json (under reports/). Candidate model: branch codex/themes-launch.

Reproduction commands (microscope):

```
python -m src.preflight # checkpoint hash, data, model, OPC UA
python -m src.segmentation.train_patches --eval # Table 8, full-section inference
python -m src.s2_section_stats # Fig. 4 and bootstrap intervals
python -m src.decision_gap --model patches --refine # decision-level results
```

Continuous integration. Four checks on the workbench branch (unit tests, offline-bundle match, local API tests and secret scanning) **[VERIFY: add the CI run URL for the final commit]**.

C Assumptions register for Section 5.4

Input	Value	Provenance
Zondereinde output	333,050 equivalent refined 4E oz (F2026)	Northam production update, 13 July 2026 [21]
Price per ounce	R32,611 (2025 average); R45,993 (H1 2026)	Valterra realised basket prices [4, 22], used as a proxy for Zondereinde
Incremental costs of recovering more metal	Not deducted	Value is revenue, not profit
Exchange rate	R17.61 per US\$	Assumed
Installed cost	US\$0.5 / 1.5 / 3.0 M	Assumed; vendor quotes required
Running cost	US\$0.1 / 0.3 / 0.6 M per year	Assumed; quotes required
Life and discount rate	5 years, 10%	Assumed